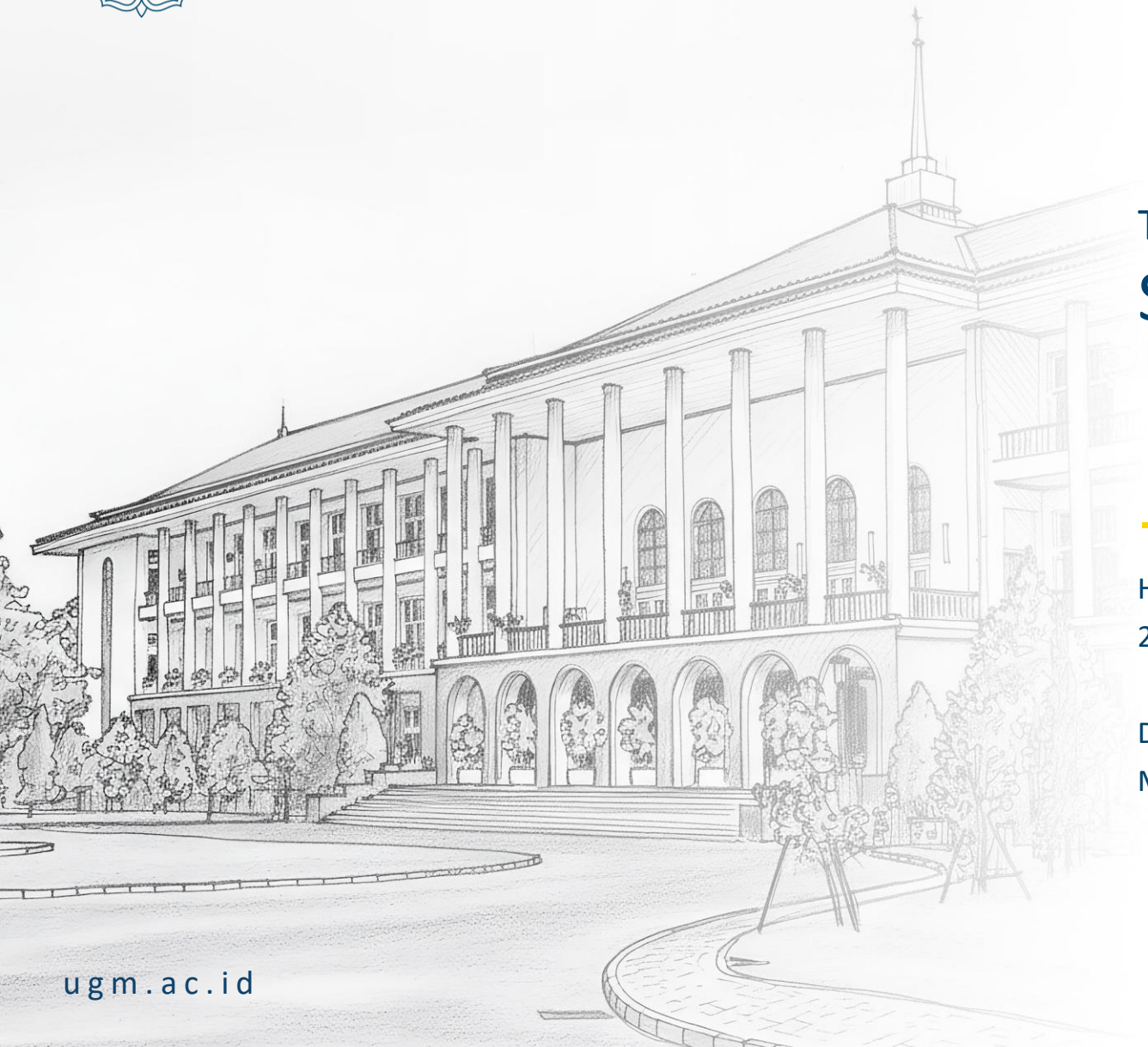
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Tugas 1

SINGLE LAYER PERCEPTRON

Hanif Affandi Hartanto
26/580088/SPA/01235

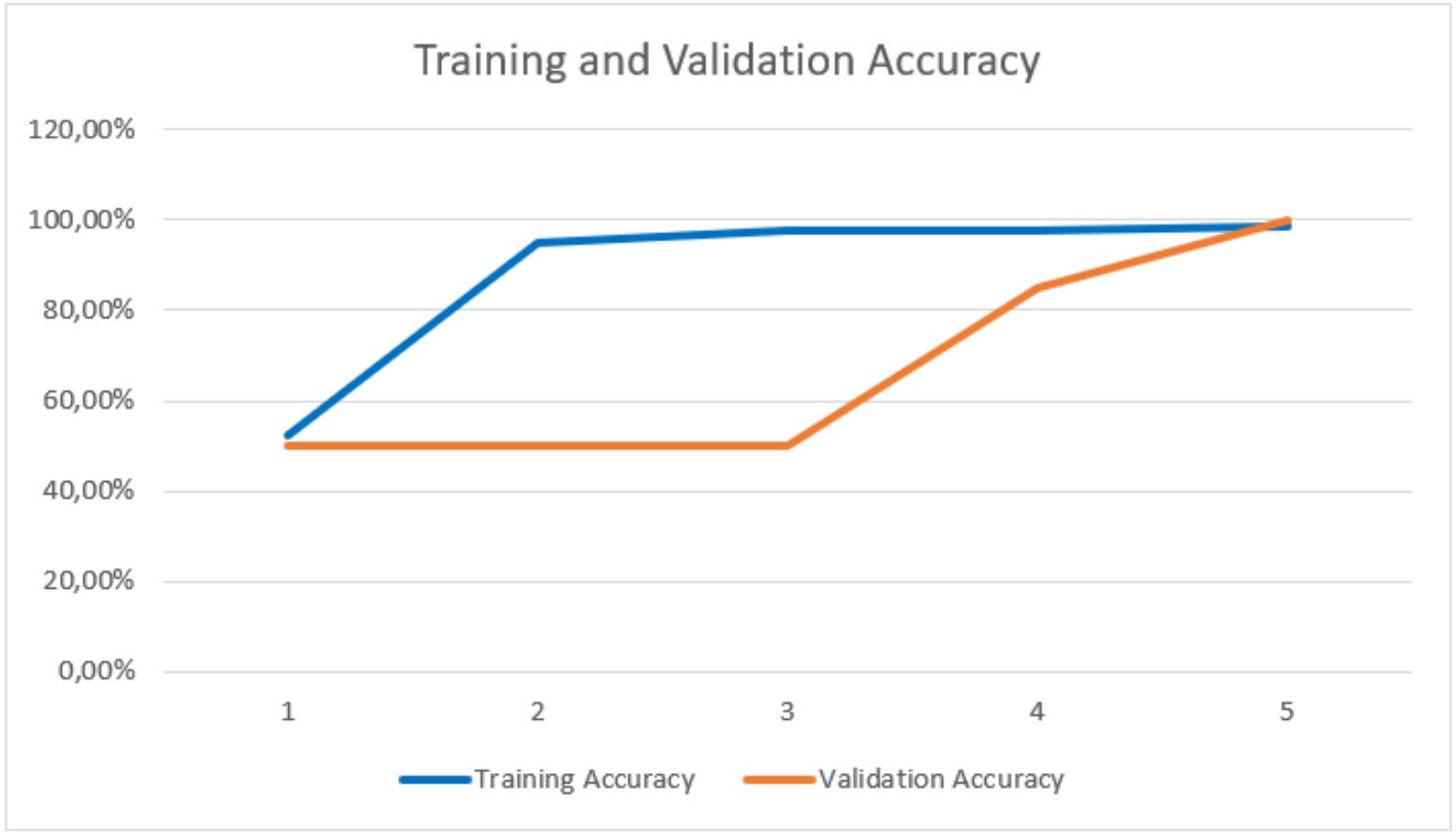
Dosen: Afiahayati, S.Kom., M.Cs., Ph.D
Mata Kuliah MII227425 Kecerdasan Buatan Lanjut Kelas B

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Training and Validation Accuracy Comparison (Google Sheets Data)

Epoch	Training Accuracy	Validation Accuracy
1	52,50%	50,00%
2	95,00%	50,00%
3	97,50%	50,00%
4	97,50%	85,00%
5	98,75%	100,00%

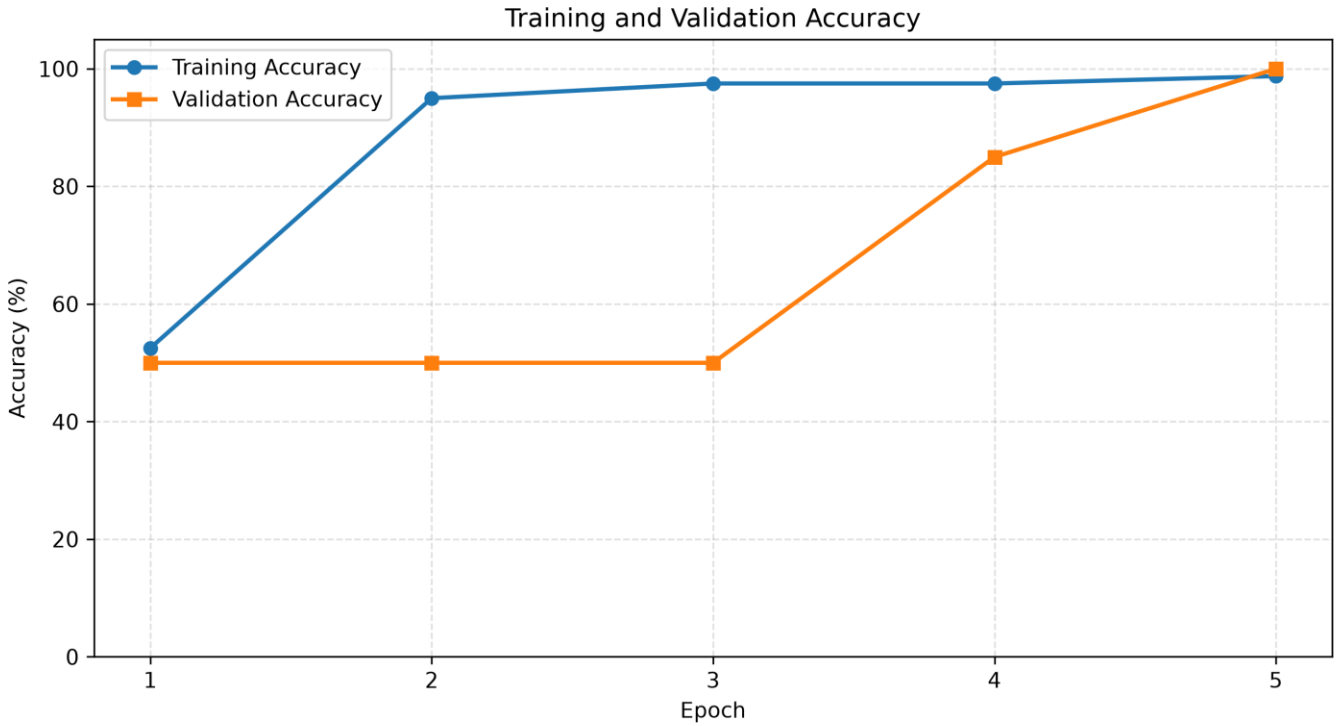


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Training and Validation Accuracy Comparison (Python Code)

```
========== SINGLE LAYER PERCEPTRON (SIGMOID) ==========
File training    : C:\Users\LENOVO\Desktop\Kecerdasan Buatan Lanjut\Tugas 1\data
File validation  : C:\Users\LENOVO\Desktop\Kecerdasan Buatan Lanjut\Tugas 1\data
Jumlah training  : 80
Jumlah validation: 20
Learning rate    : 0.1
Bobot awal       : [0.5, 0.5, 0.5, 0.5, 0.5]
Epoch            : 5

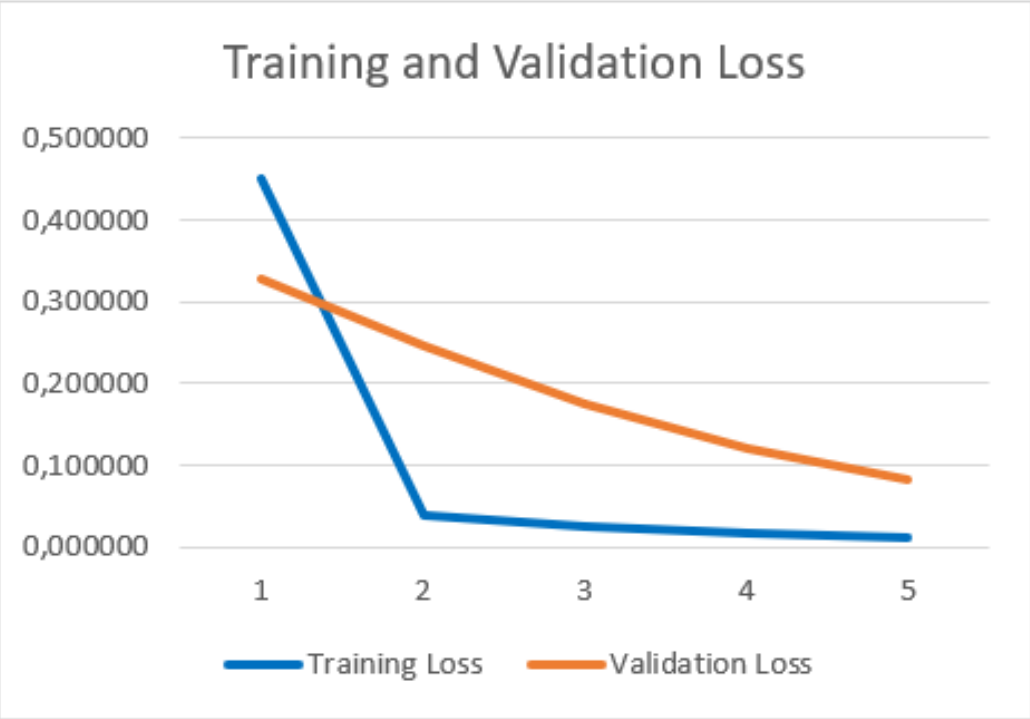
Epoch | Train Accuracy | Train Loss | Validation Accuracy | Validation Loss
-------------------------------------------------------------------------------
    1 |         52.50% |  0.449889 |              50.00% |       0.328951
    2 |         95.00% |  0.037452 |              50.00% |       0.247289
    3 |         97.50% |  0.024372 |              50.00% |       0.175892
    4 |         97.50% |  0.017357 |              85.00% |       0.119381
    5 |         98.75% |  0.012740 |             100.00% |       0.081581
```



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Training and Validation Loss Comparison (Google Sheets Data)

Epoch	Training Loss	Validation Loss
1	0,449889	0,328951
2	0,037452	0,247289
3	0,024372	0,175892
4	0,017357	0,119381
5	0,012740	0,081581

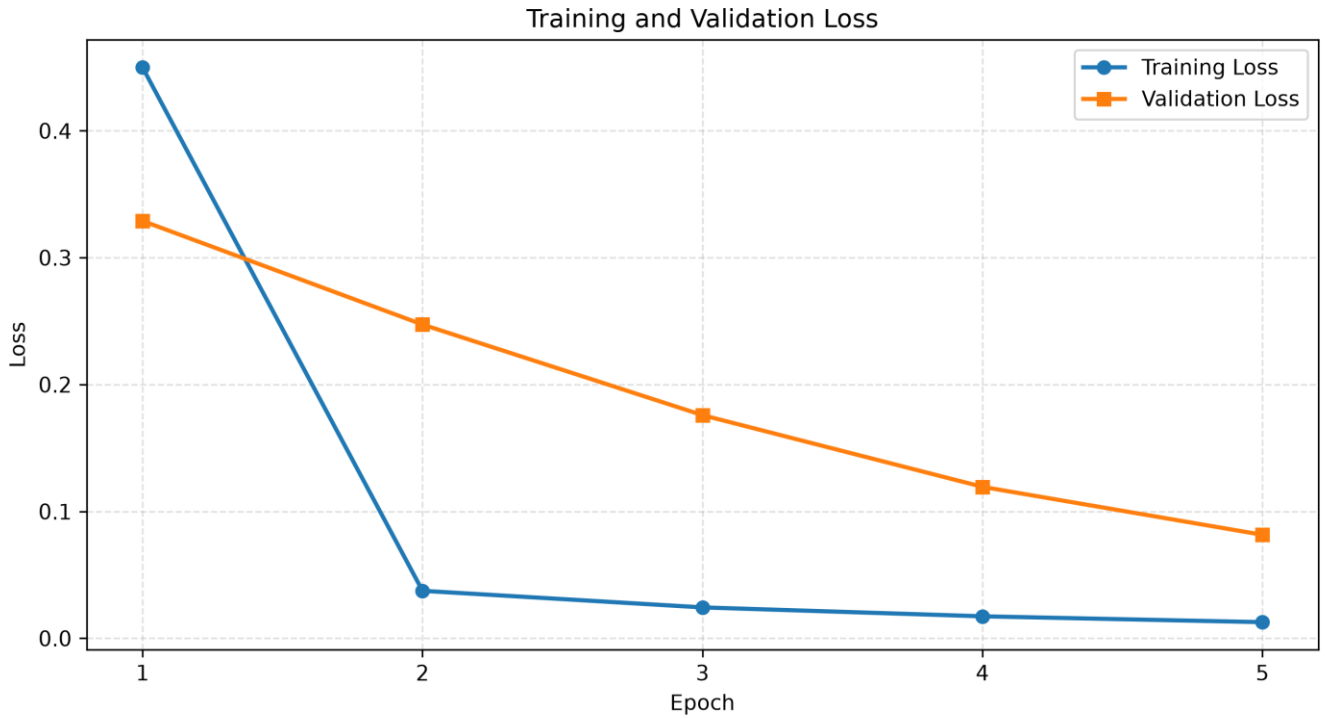


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Training and Validation Loss Comparison (Python Code)

```
========== SINGLE LAYER PERCEPTRON (SIGMOID) ==========
File training    : C:\Users\LENOVO\Desktop\Kecerdasan Buatan Lanjut\Tugas 1\data
File validation  : C:\Users\LENOVO\Desktop\Kecerdasan Buatan Lanjut\Tugas 1\data
Jumlah training  : 80
Jumlah validation: 20
Learning rate    : 0.1
Bobot awal       : [0.5, 0.5, 0.5, 0.5, 0.5]
Epoch            : 5

Epoch | Train Accuracy | Train Loss | Validation Accuracy | Validation Loss
------------------------------------------------------------------------------
    1 |         52.50% |  0.449889 |              50.00% |       0.328951
    2 |         95.00% |  0.037452 |              50.00% |       0.247289
    3 |         97.50% |  0.024372 |              50.00% |       0.175892
    4 |         97.50% |  0.017357 |              85.00% |       0.119381
    5 |         98.75% |  0.012740 |             100.00% |       0.081581
```



URL PROJECT

Google Sheets:

https://docs.google.com/spreadsheets/d/105LuH365fYt2LnLFiJJ6xjjr8vDJM6pm

GitHub Repository:

https://github.com/hanifaffandihartanto/SingleLayerPerceptron

REFERENSI

Bahan Materi Kuliah terkait Single Layer Perceptron (SLP):

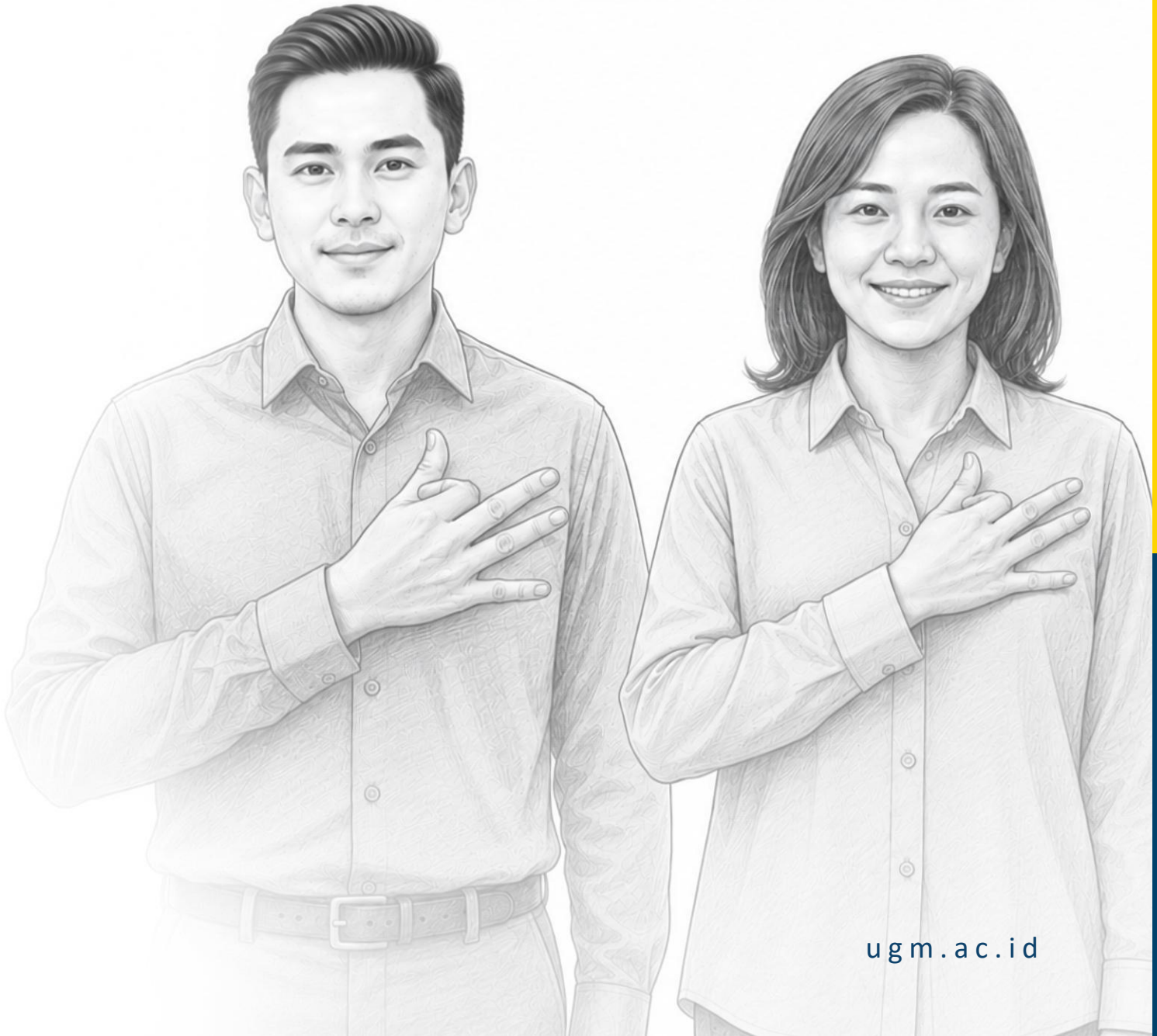
https://drive.google.com/drive/folders/1mRsriHV10VlcOX6YJACCTiAavkXJOSe1

PMM-TemplateSLP

https://docs.google.com/spreadsheets/d/1yhi8ASebpmljJ3NpYQ7NmxlWLmhGOI6W

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THANK YOU



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